Inheritance II: Mendelian Traits

Dr. Laci Gerhart

Intro Bio: Ecology and Evolution

Opening Question

You cross a purple-flowered plant with a white-flowered plant. Assuming there are no mutations, what color flowers will the offspring have?

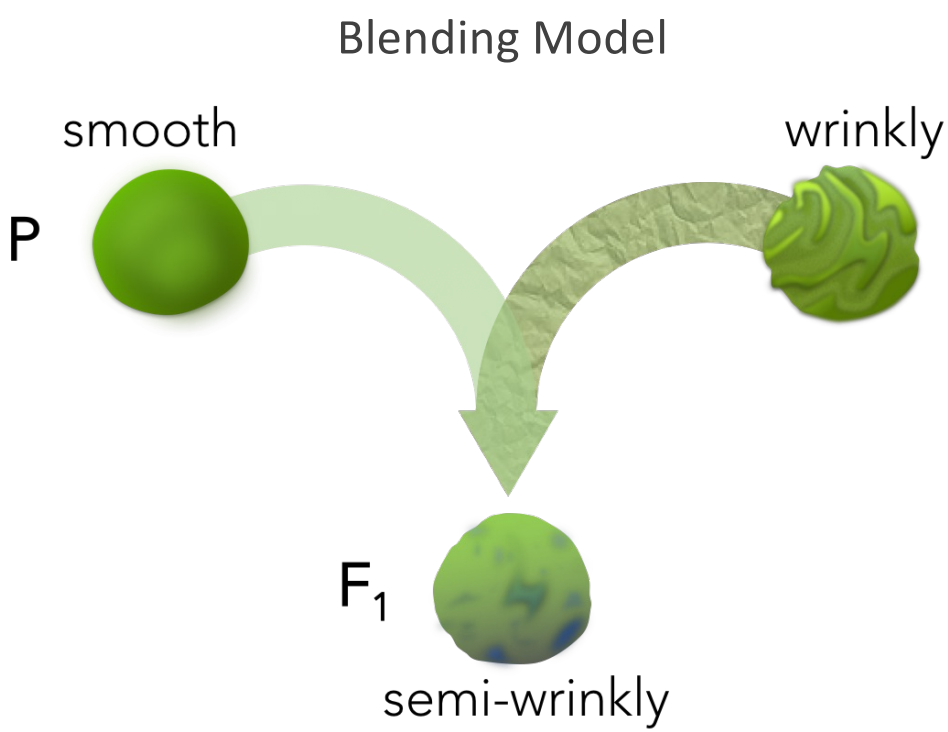
A. Purple
B. White
C. Lavender
D. Purple and white splotches or stripes
E. Any of the above?

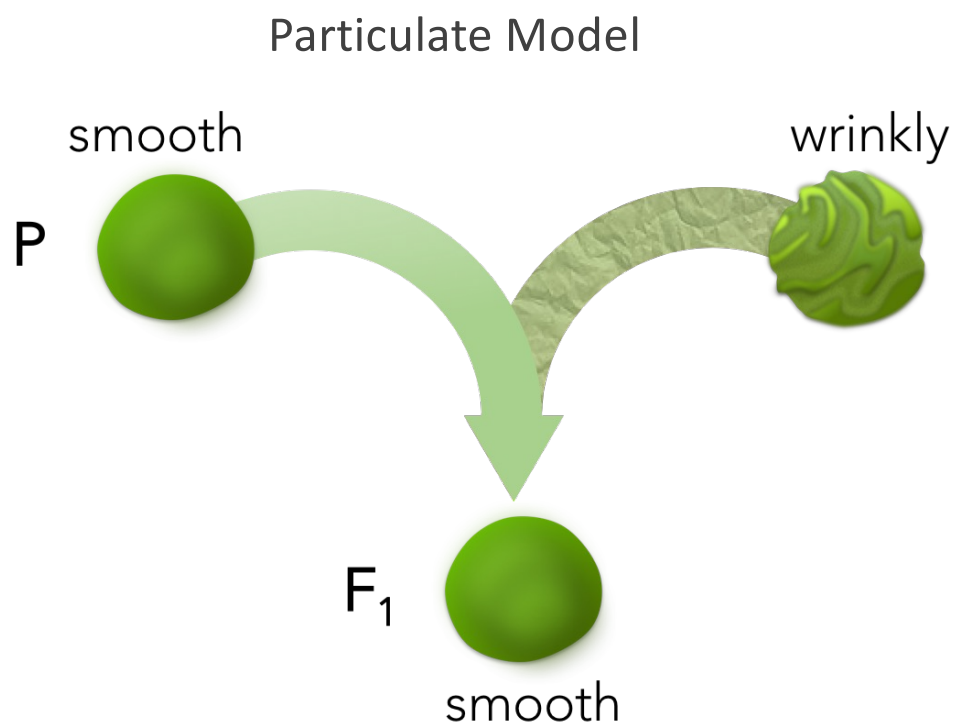
Learning Objectives

By the end of this lecture, students should be able to…

- Accurately use genetic terms such as genotype, phenotype, allele, homozygous, heterozygous, etc
- Use the multiplication rule to calculate the joint probabilities resulting from independent assortment
- Explain the relationship between Mendel's laws and meiotic division
- Predict the expected ratios of genotypes and phenotypes of offspring of a particular mating (Punnett square)
- Predict allele interactions under different forms of dominance (simple/complete, incomplete, codominant)

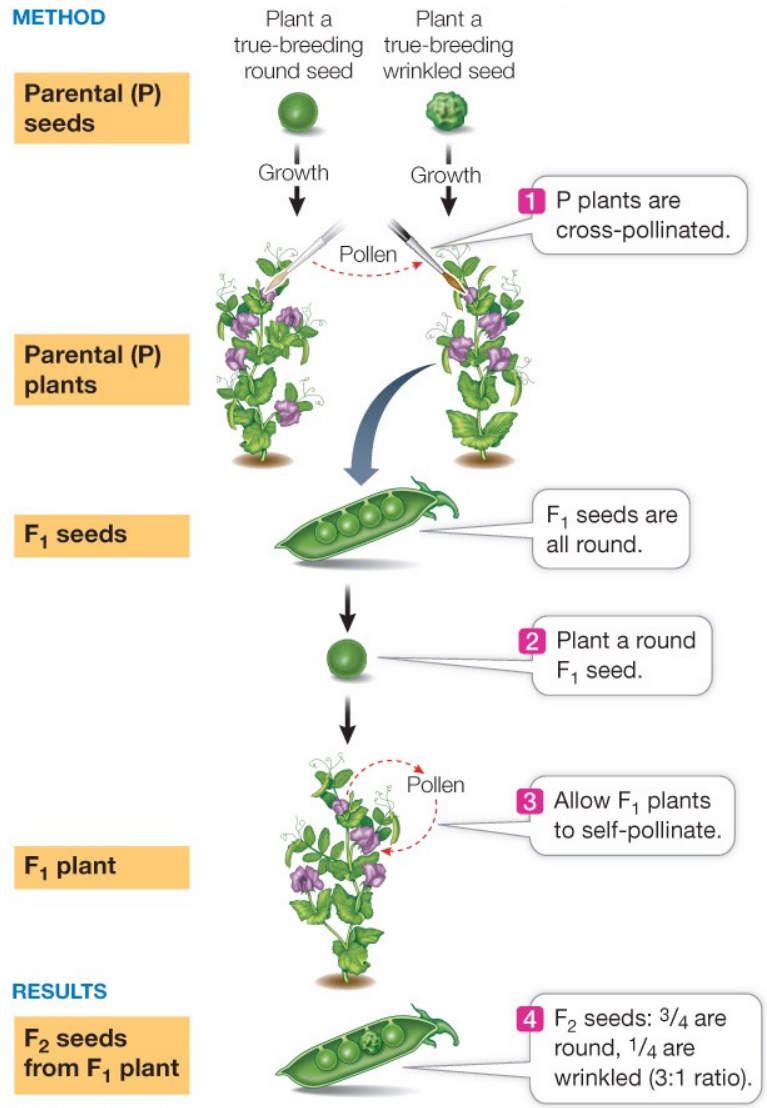
Models of Inheritance in the 1800s





Mendel's Pea Experiments

Hypothesis:
when two strains of peas with contrasting traits are bred, their characteristics are irreversibly blended in succeeding generations.

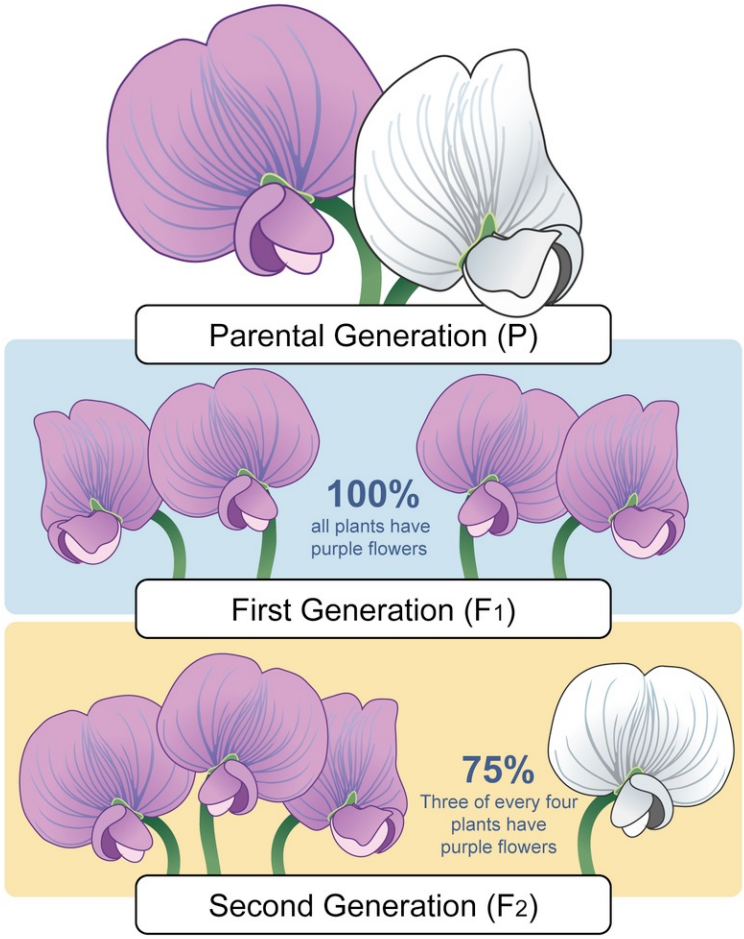


Did Mendel accept or reject his hypothesis?

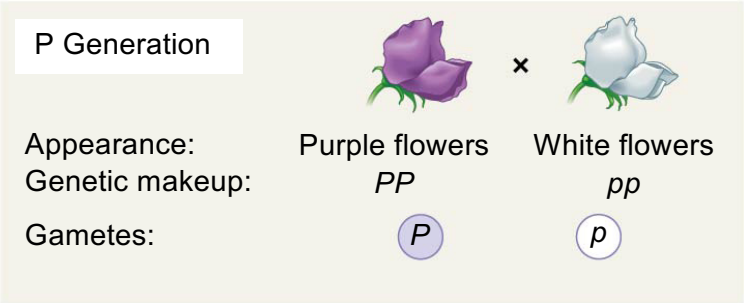
A. Accept B. Reject C. ???

Mendel's Pea Traits

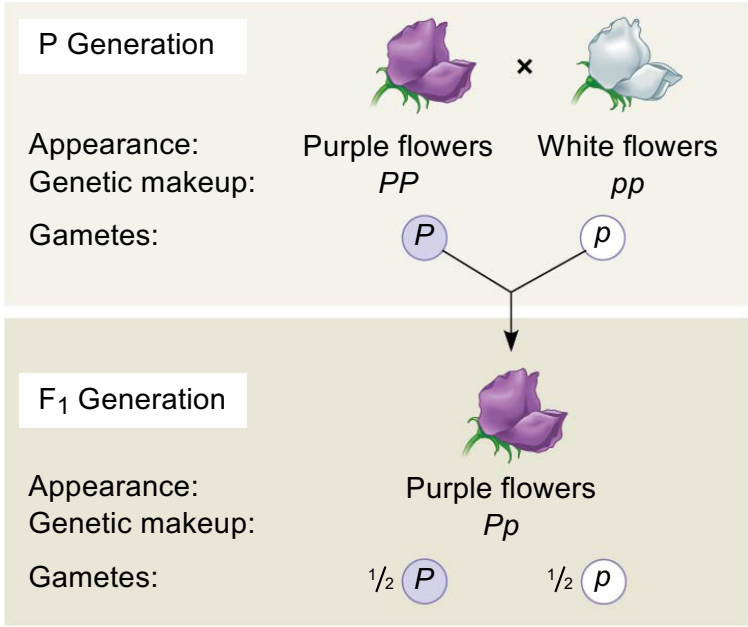
Parental generation phenotypes		
Dominant		Recessive
Round seeds	×	Wrinkled seeds
Yellow seeds	×	Green seeds
Purple flowers	×	White flowers
Inflated pods	×	Constricted pods
Green pods	×	Yellow pods
Axial flowers	×	Terminal flowers
Tall stems (1 m)	×	Dwarf stems (0.3 m)



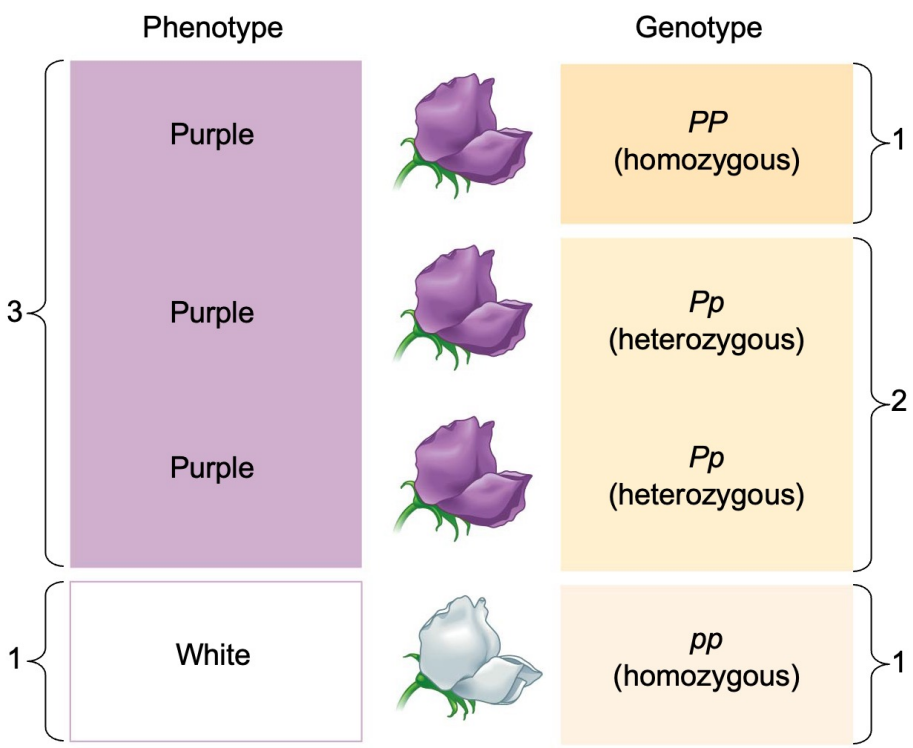
Crosses

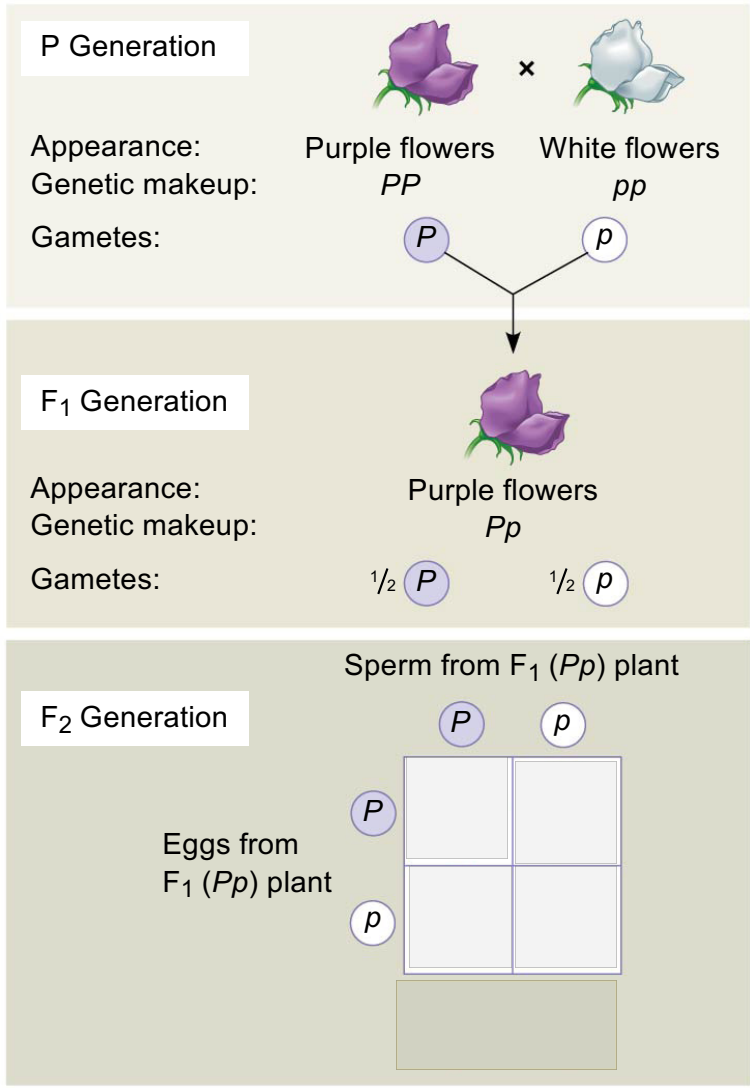

P Generation
×
Appearance:
Genetic makeup:
Gametes:
Purple flowers
PP
P
White flowers
pp
p

Crosses


P Generation
×
Appearance:
Genetic makeup:
Purple flowers
PP
White flowers
pp
Gametes:
P
p
F1 Generation
Appearance:
Genetic makeup:
Purple flowers
Pp
Gametes:
1/2 P
1/2 p

Crosses


Phenotype
Genotype
Purple
Purple
Purple
White
3
1
PP
(homozygous)
Pp
(heterozygous)
Pp
(heterozygous)
pp
(homozygous)
1
2
1


P Generation
×
Appearance:
Genetic makeup:
Purple flowers
PP
White flowers
pp
Gametes:
P
p
F1 Generation
Appearance:
Genetic makeup:
Purple flowers
Pp
Gametes:
1/2 P
1/2 p
F2 Generation
Sperm from F1 (Pp) plant
P
p
Eggs from
F1 (Pp) plant
P
p

Mendel's Pea Traits

All of these traits are single-gene, with only two alleles, showing simple/complete dominance

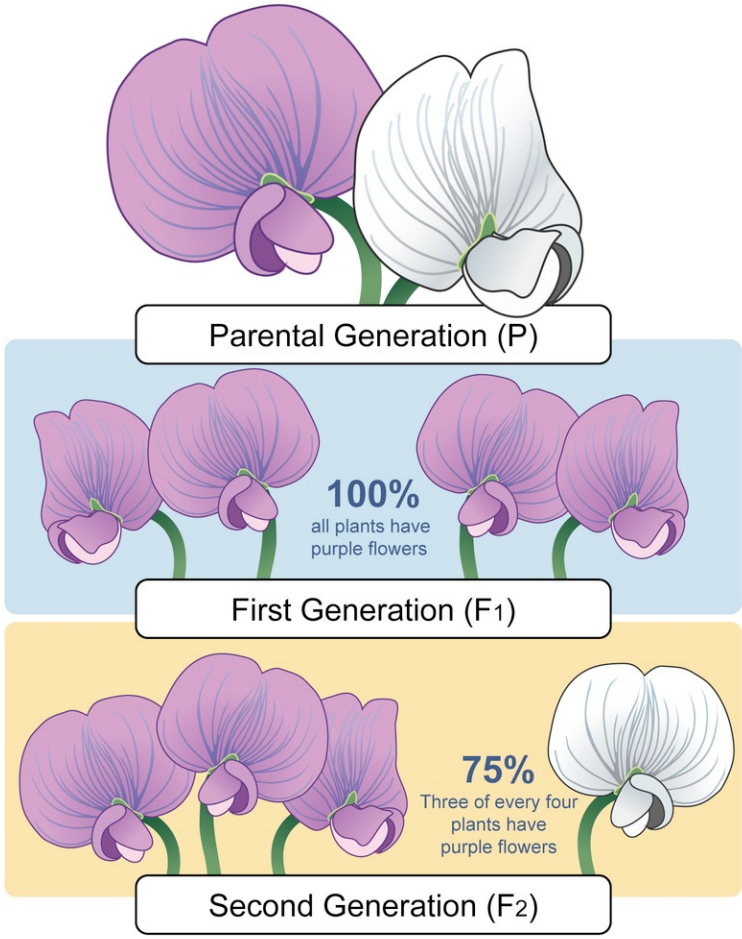
	Parental generation phenotypes			F_2 generation phenotypes			
	Dominant		Recessive	Dominant	Recessive	Total	Ratio
	Round seeds	×	Wrinkled seeds	5,474	1,850	7,324	2.96:1
	Yellow seeds	×	Green seeds	6,022	2,001	8,023	3.01:1
	Purple flowers	×	White flowers	705	224	929	3.15:1
	Inflated pods	×	Constricted pods	882	299	1,181	2.95:1
	Green pods	×	Yellow pods	428	152	580	2.82:1
	Axial flowers	×	Terminal flowers	651	207	858	3.14:1
	Tall stems (1 m)	×	Dwarf stems (0.3 m)	787	277	1,064	2.84:1

Types of Dominance

- **Simple/Complete Dominant:** a single dominant allele produces the dominant phenotype. The homozygous dominant and heterozygous genotypes have the same phenotype.

- **Incomplete Dominant:** the heterozygote phenotype is intermediate between the two homozygous phenotypes

- **Codominant:** the heterozygote phenotype is both of the homozygous phenotypes expressed fully



Types of Dominance

The homozygous dominant phenotype is red and the homozygous recessive phenotype is white.

What will the heterozygote look like if this trait exhibits:

- Complete Dominance?
- Incomplete Dominance?
- Codominance?

RR

rr

Opening Question

You cross a purple-flowered plant with a white-flowered plant. Assuming there are no mutations, what color flowers will the offspring have?

A. Purple
B. White
C. Lavender
D. Purple and white splotches or stripes
E. Any of the above?

Mendel's Laws

1822-1884

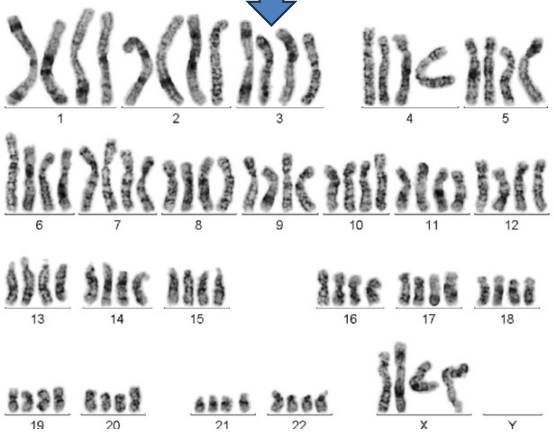
- **Law of Segregation:** when any individual produces gametes, the two copies of a gene separate so that each gamete receives only one copy

- **Law of Independent Assortment:** alleles of different genes assort independently of one another during gamete formation

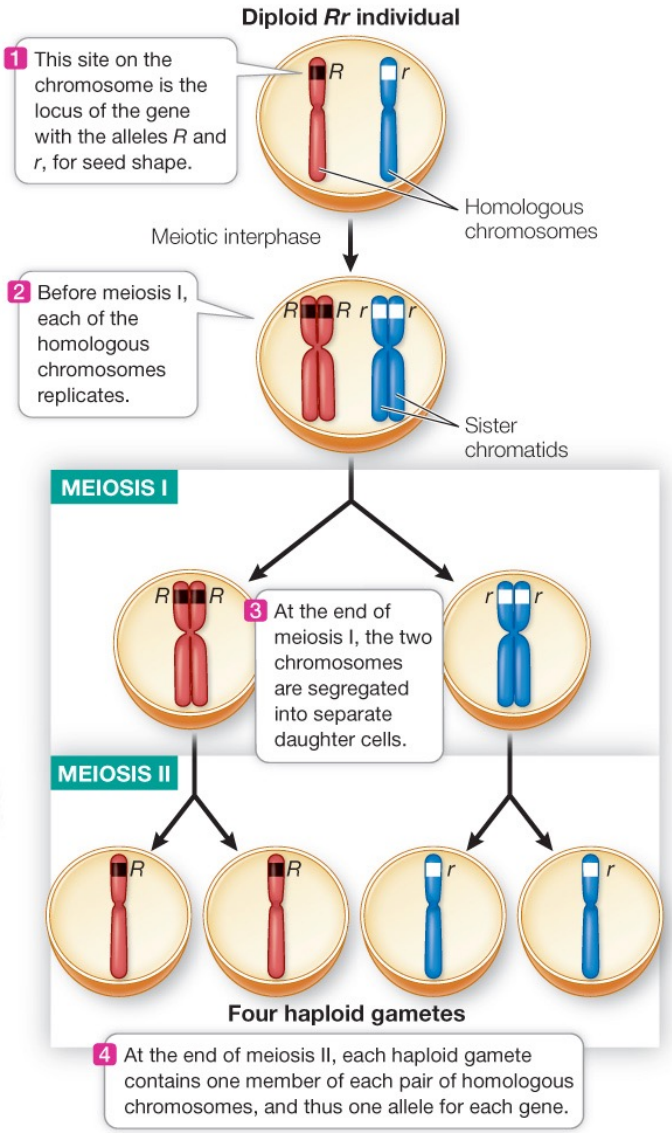
Law of Segregation

When an individual produces gametes, the two copies of a gene separate, so that each gamete receives only one copy

If segregation didn't occur, the number of chromosomes would double each generation

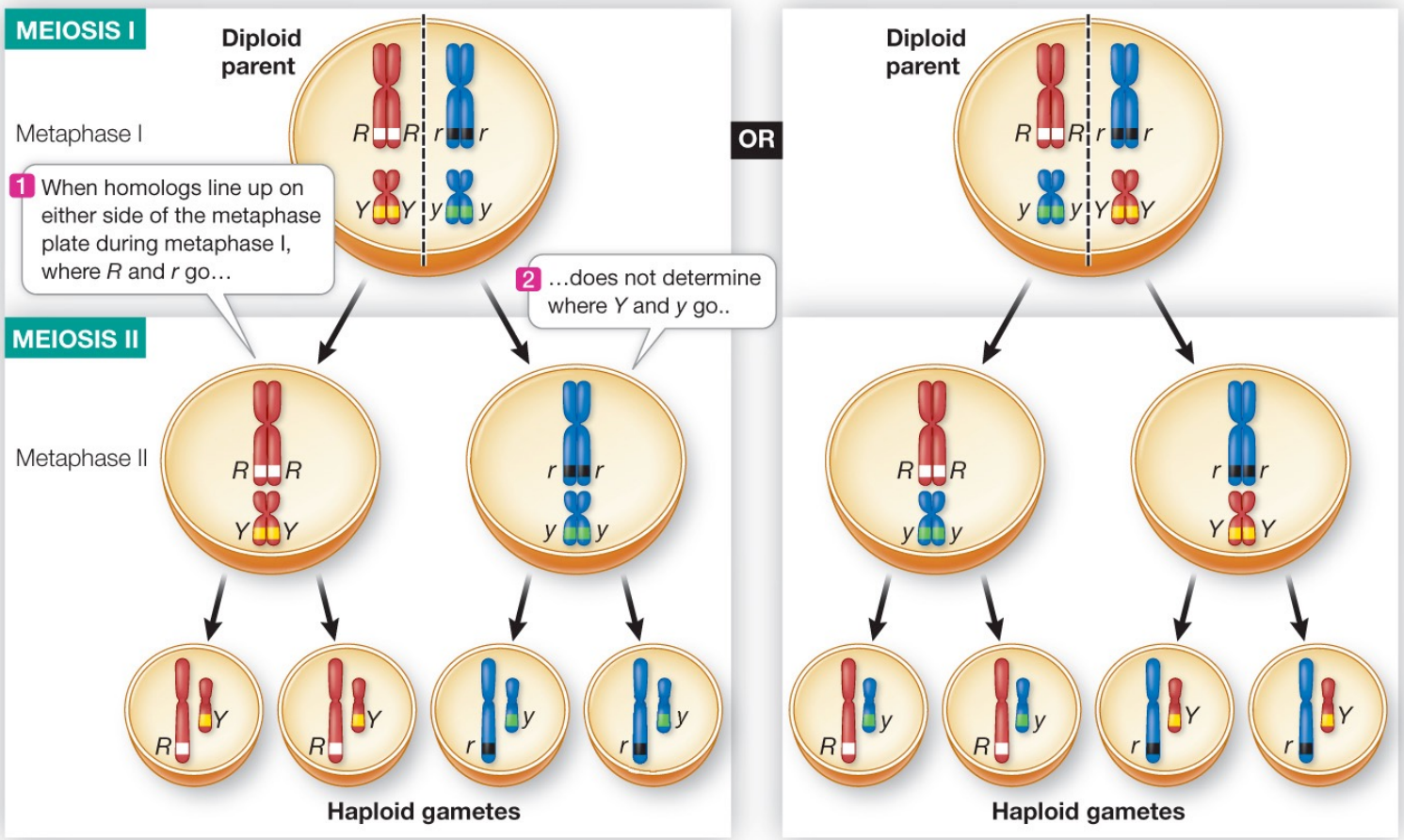


Segregation allows the number of chromosomes to stay constant (2n)



Law of Independent Assortment

Alleles of different genes sort independently of one another during gamete formation.

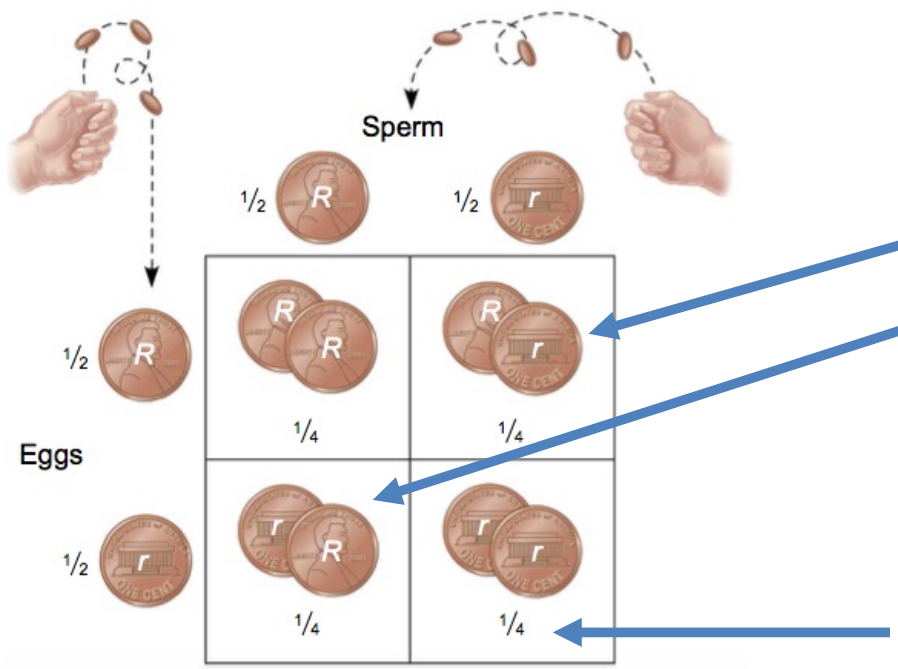


Discussion 5: Dance of the Meiotic Division

Research Questions:

- How do independent assortment and segregation of chromosomes during metaphase affect the resulting gamete genotypes?
- Do actual gamete frequencies in your class match Mendelian predictions?

Calculating Probabilities



<u>The multiplication and addition rules</u>

AND: multiply the probabilities

OR: add the probabilities

The probability of being a heterozygote:

P(R sperm) AND P(r egg) = 0.5 x 0.5 = 0.25

OR

P(r sperm) AND P(R egg) = 0.5 x 0.5 = 0.25

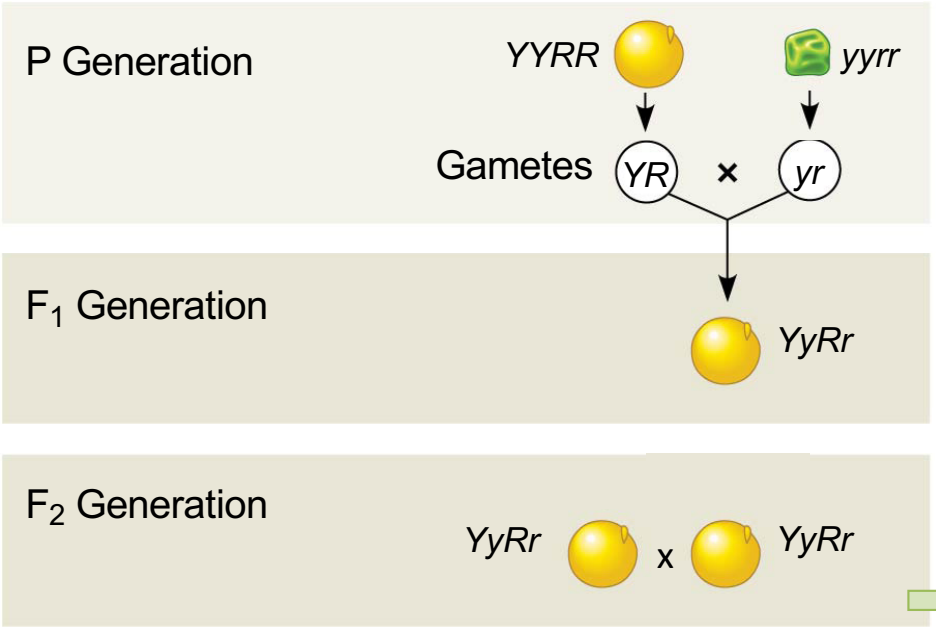
0.25 + 0.25 = 0.5

The probability of inheriting an r from the sperm AND an r from the egg

P(r sperm) AND P(r egg) = 0.5 x 0.5 = 0.25

Considering Multiple Genes

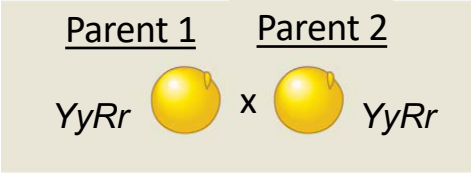
Color: Y = yellow, y = green
Shape: R = round, r = wrinkled



gametes from heterozygous parent	YR	yR	Yr	yr
YR				
yR				
Yr				
yr				

Calculating Probabilities

Color: Y = yellow, y = green
Shape: R = round, r = wrinkled

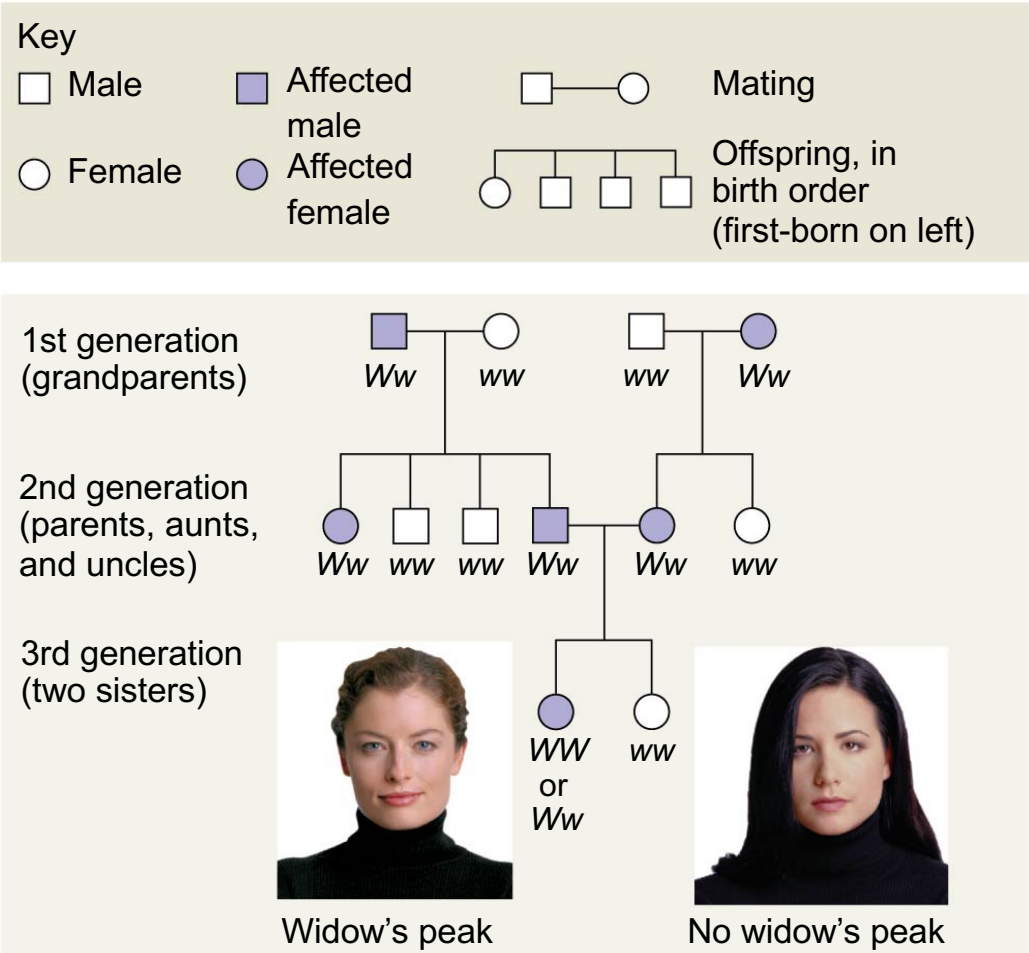


- What is the probability of Parent 1 producing a *Yr* gamete?

- What is the probability that an offspring receives a *Yr* gamete from Parent 1 and a *Yr* gamete from Parent 2?

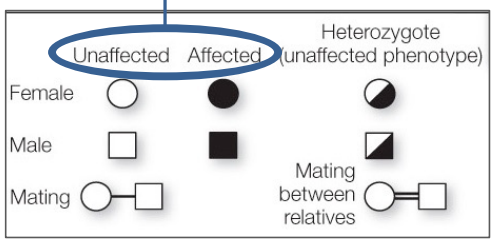
- What is the probability that the offspring has the phenotype of yellow and wrinkled?

Pedigrees: Tracking Phenotypes Across Generations

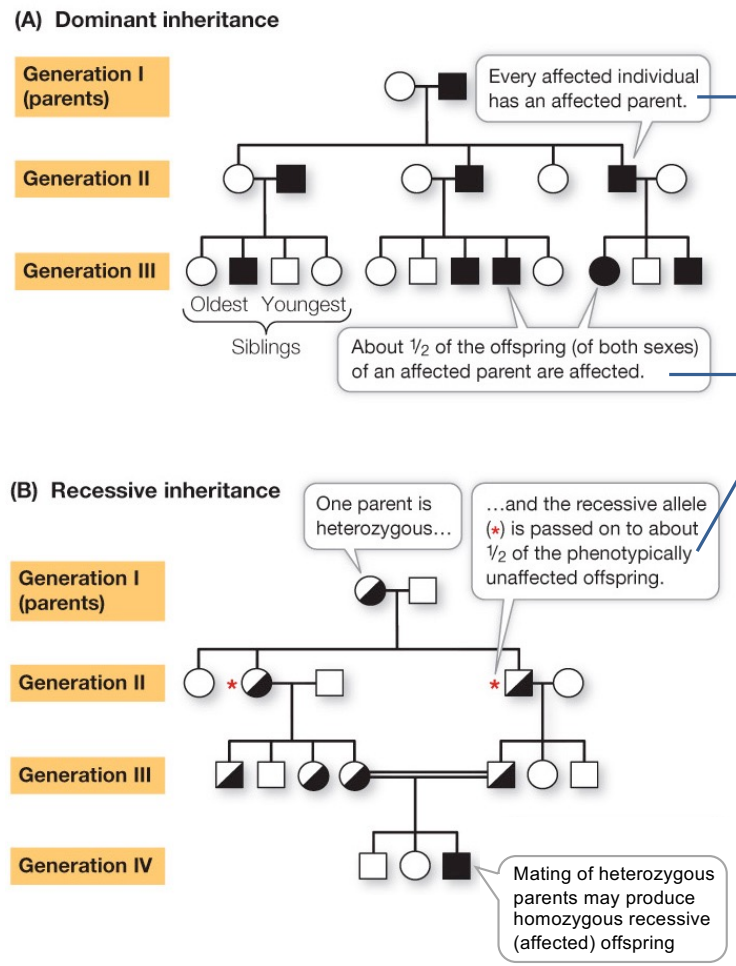


Interpreting Pedigrees

This language only makes sense for disease phenotypes



There is one parent/offspring phenotype pattern that reliably indicates which phenotype is dominant and which is recessive



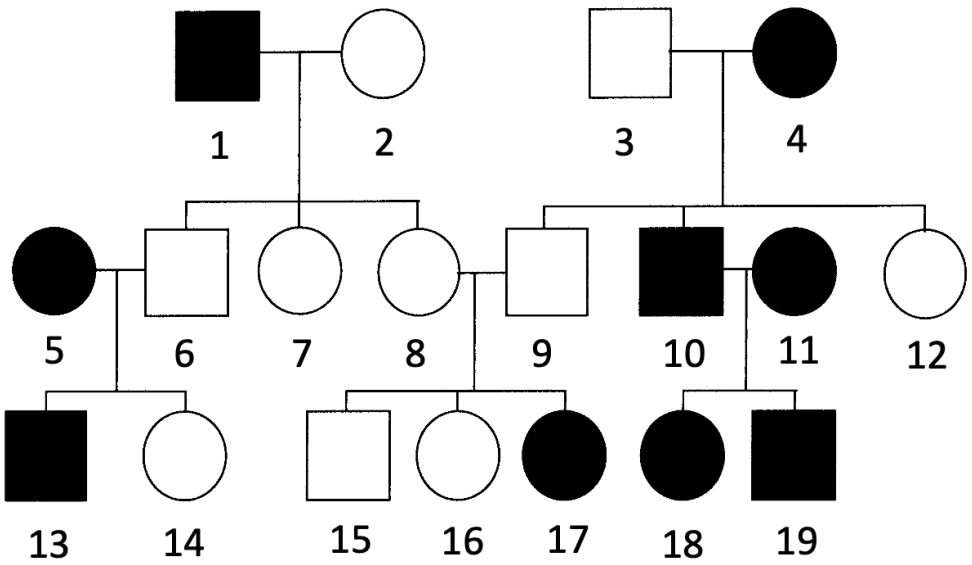
Can also be true of the recessive phenotype

Human offspring numbers are far too small to rely on Mendelian expectations

Discussion Question

The pedigree below shows the inheritance pattern for a simple dominant trait with two alleles. Which phenotype is dominant?

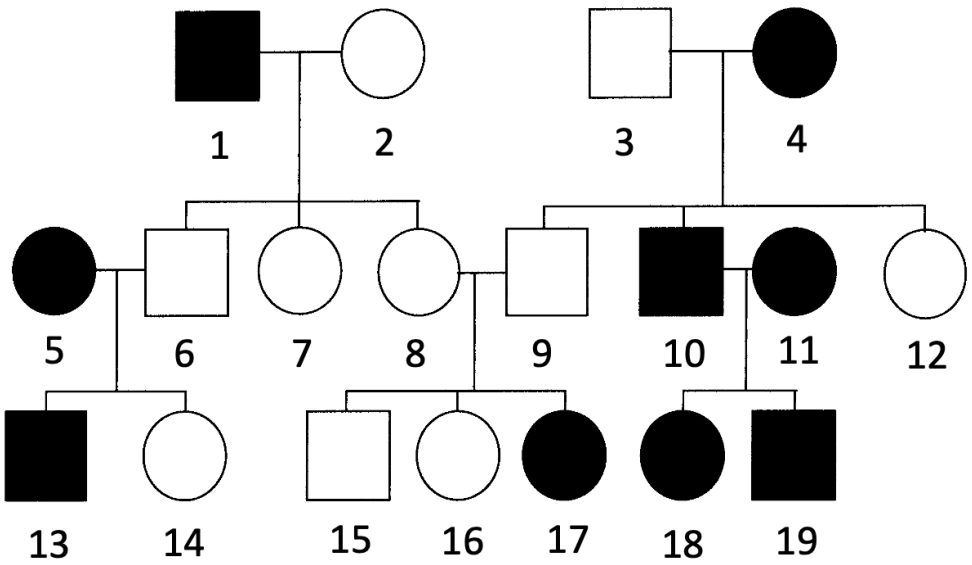
A. Filled shapes
B. Open shapes
C. ….one of the two?



Discussion Question

What is the genotype of Individual 7?

A. Homozygous dominant
B. Homozygous recessive
C. Heterozygous
D. Either homozygous dom or heterozygous, we can't be certain

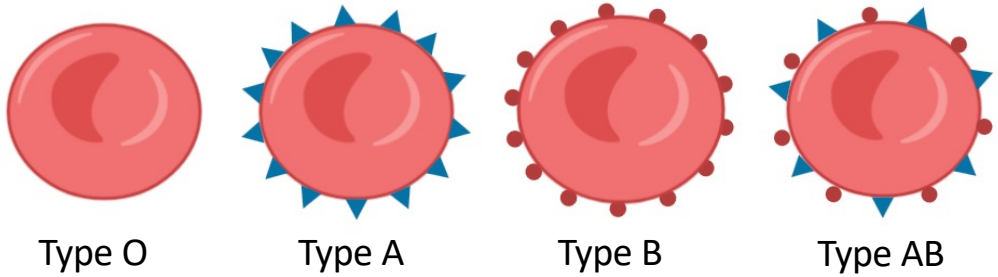


Discussion Question

Humans have three alleles for blood type, *A*, *B,* and *O*. Possible genotypes and their phenotypes are shown in the table at right.

What type of dominance do these alleles show?

A. Complete dominance
B. Incomplete dominance
C. Codominance
D. More than one type of dominance?



Genotype	Phenotype (blood type)
AA	Type A
AO	Type A
BB	Type B
BO	Type B
OO	Type O
AB	Type AB

Discussion Question

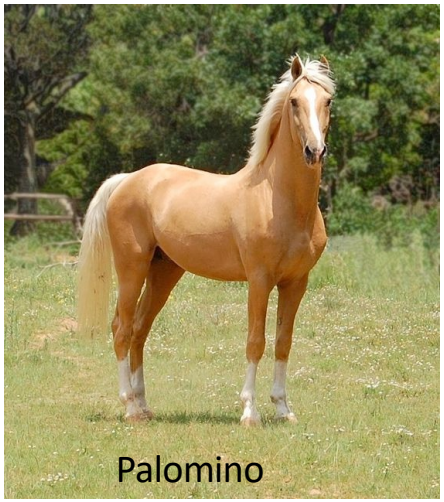
Two parents believe their baby was switched at the hospital with someone else's child. The mother has blood type O+, the father has blood type AB-, and the baby has blood type B+. Are the parents correct?

Genotype	Phenotype
AA	Type A
AO	Type A
BB	Type B
BO	Type B
OO	Type O
AB	Type AB

Genotype	Phenotype
+/+	Positive
+/-	Positive
-/-	Negative

What type of dominance do these alleles show?

Discussion Question

In a fictional population of horses, brown color is produced by the genotype *BB*, white coat color is produced by the genotype *bb* and heterozygotes (*Bb*) are a yellowish-tan color called palomino.

What color of parent horses should you breed in order to obtain the most palominos?